1. a) Derivar una definición recursiva para la función especificada por:

$$f.xs = \left\langle \forall i : 0 \le i < \#xs : (-1)^{xs\,!\,i} \ge 0 \right\rangle$$

a donde xs es una lista de números naturales (≥ 0).

b) Dar una lista xs de al menos 3 elementos que cumpla $f.xs = True$. Justificar.

c) Dar otra de al menos 3 elementos que cumpla $f.xs = False$. Justificar.

a)

$f.xs = \langle \forall i : 0 \le i < \#xs : (-1)^{xs!i} \ge 0 \rangle$ (H)

CB) $f.[\,] = \langle \forall i : 0 \le i < \#[\,] : (-1)^{[\,]!i} \ge 0 \rangle$

$= \{$ Def length, logica $\}$

$\langle \forall i : false : (-1)^{[\,]!i} \ge 0 \rangle$

$= \{$ Rango vacio $\}$

$\boxed{True}$

CR) $f.(x \triangleright xs) = \langle \forall i : 0 \le i < \#(x \triangleright xs) : (-1)^{(x \triangleright xs)!i} \ge 0 \rangle$

$= \{$ logica $\}$

$\langle \forall i : i = 0 \lor 1 \le i < \#(x \triangleright xs) : (-1)^{(x \triangleright xs)!i} \ge 0 \rangle$

$= \{$ Partición de rango $\}$

$\langle \forall i : i = 0 : (-1)^{(x \triangleright xs)!i} \ge 0 \rangle \land \langle \forall i : 1 \le i < \#(x \triangleright xs) : (-1)^{(x \triangleright xs)!i} \ge 0 \rangle$

$= \{$ Rango unitario $\}$

$(-1)^{(x \triangleright xs)!0} \ge 0 \land \langle \forall i : 1 \le i < \#(x \triangleright xs) : (-1)^{(x \triangleright xs)!i} \ge 0 \rangle$

$= \{$ Def index $\}$

$(-1)^{x} \ge 0 \land \langle \forall i : 1 \le i < \#(x \triangleright xs) : (-1)^{(x \triangleright xs)!i} \ge 0 \rangle$

$= \{$ Def length, cambio de variable $i := i+1$, aritmetica $\}$

$(-1)^{x} \ge 0 \land \langle \forall i : 0 \le i < \#xs : (-1)^{(x \triangleright xs)!(i+1)} \ge 0 \rangle$

$= \{$ Def index $\}$

$(-1)^{x} \ge 0 \land \langle \forall i : 0 \le i < \#xs : (-1)^{xs!i} \ge 0 \rangle$

$= \{$ HI $\}$

$(-1)^{x} \ge 0 \land f.xs$

$f : [Num] \to Bool$

$f.[\,] \doteq true$

$f.(x \triangleright xs) \doteq (-1)^{x} \ge 0 \land f.xs$

b) $f.[2,4,6] = \langle \forall i : 0 \le i < \#[2,4,6] : (-1)^{[2,4,6]!i} \ge 0 \rangle$

$= \{$ Def de length, logica $\}$

$\langle \forall i : i \in \{1,2,3\} : (-1)^{[2,4,6]!i} \ge 0 \rangle$

$= \{$ Aplicación del rango al termino $\}$

$(-1)^{[2,4,6]!1} \ge 0 \land (-1)^{[2,4,6]!2} \ge 0 \land (-1)^{[2,4,6]!3} \ge 0$

$= \{$ Def index

$(-1)^{2} \ge 0 \land (-1)^{4} \ge 0 \land (-1)^{6} \ge 0$

$= \{$ Aritmetica $\}$

$1 \ge 0 \land 1 \ge 0 \land 1 \ge 0$

$= \{$ Desigualdad y logica $\}$

$\boxed{True}$

c) $f.[1,3,5] = \langle \forall i : 0 \leq i < \#[2,4,6] : (-1)^{[1,3,5]!i} \geq 0 \rangle$

$= \{$ Def de length, logica $\}$

$\langle \forall i : i \in \{1,2,3\} : (-1)^{[1,3,5]!i} \geq 0 \rangle$

$= \{$ Aplicación del rango al termino $\}$

$(-1)^{[1,3,5]!1} \geq 0 \wedge (-1)^{[1,3,5]!2} \geq 0 \wedge (-1)^{[1,3,5]!3} \geq 0$

$\{$ Def index

$(-1)^{2} \geq 0 \wedge (-1)^{3} \geq 0 \wedge (-1)^{5} \geq 0$

$\{$ Aritmetica $\}$

$-1 \geq 0 \wedge -1 \geq 0 \wedge -3 \geq 0$

$= \{$ Desigualdad y logica $\}$

$\boxed{False}$

2. a) Derivar una definición recursiva para la función especificada por:

$$pin8.xs = \langle \exists as, bs \ : xs = as ++ bs : \ prod.as = 8 \rangle$$

a donde *prod* es la función que calcula el producto de los elementos de una lista.

b) Dar una lista xs de al menos 3 elementos que cumpla $pin8.xs = True$. Justificar.

c) Dar otra de al menos 3 elementos que cumpla $pin8.xs = False$. Justificar.

a) $Pin8.xs = \langle \exists as, bs : xs = as ++ bs : Prod.as = 8 \rangle$

CB $Pin8.[\,] = \langle \exists as, bs : [\,] = as ++ bs : Prod.as = 8 \rangle$

$= \{$ Prop de concatenar $\}$

$\langle \exists as, bs : as = [\,] \wedge bs = [\,] : Prod.as = 8 \rangle$

$= \{$ eliminación de variable bs, Rango unitario $\}$

$Prod.[\,] = 8$

$= \{$ Def de Prod $\}$

$1 = 8$

$= \{$ logica $\}$

$\boxed{False}$

CR $Pin.(x \triangleright xs) = \langle \exists as, bs : (x \triangleright xs) = as ++ bs : Prod.as = 8 \rangle$

$= \{$ logica $\}$

$\langle \exists as, bs : (x \triangleright xs) = as ++ bs \wedge (as = [\,] \vee as \neq [\,]) : Prod.as = 8 \rangle$

$= \{$ Distributividad, Partición de rango $\}$

$\langle \exists as, bs : (x \triangleright xs) = as ++ bs \wedge as = [\,] : Prod.as = 8 \rangle \vee \langle \exists as, bs : (x \triangleright xs) = as ++ bs \wedge as \neq [\,] : Prod.as = 8 \rangle$

$= \{$ eliminación de variable,

$\langle \exists as, bs : (x \triangleright xs) = bs : Prod.[\,] = 8 \rangle \vee \langle \exists as, bs : (x \triangleright xs) = as ++ bs \wedge as \neq [\,] : Prod.as = 8 \rangle$

$= \{$ Def Prod, logica $\}$

$False \vee \langle \exists as, bs : (x \triangleright xs) = as ++ bs \wedge as \neq [\,] : Prod.as = 8 \rangle$

$= \{$ Cambio de variable $as := (a \triangleright as) \}$

$\langle \exists as, bs : (x \triangleright xs) = (a \triangleright as) +\!+\, bs : Prod.(a \triangleright as) = 8 \rangle$

$= \{$ Def concatenar $\}$

$\langle \exists as, bs : (x \triangleright xs) = a \triangleright (as +\!+\, bs) : Prod.(a \triangleright as) = 8 \rangle$

$= \{$ Prop de listas $\}$

$\langle \exists as, bs : x = a \wedge xs = as +\!+\, bs : Prod.(a \triangleright as) = 8 \rangle$

$= \{$ eliminación de variable $\}$

$\langle \exists as, bs : xs = as +\!+\, bs : Prod.(x \triangleright as) = 8 \rangle$

$= \{$ Def de Prod $\}$

$\langle \exists as, bs : xs = as +\!+\, bs : x \cdot Prod.as = 8 \rangle$

$= \{$ generalizo $\}$

$gPin8.n.xs = \langle \exists as\, bs : xs = as +\!+\, bs : n \cdot Prod.as = 8 \rangle$ (H)

(CB) $gPin8.n.[\,] = \langle \exists as, bs : [\,] = as +\!+\, bs : n \cdot Prod.as = 8 \rangle$

$= \{$ Prop de listas $\}$

$\langle \exists as, bs : as = [\,] \wedge bs = [\,] : n \cdot Prod.as = 8 \rangle$

$= \{$ eliminación de var bs, Rango unitario $\}$

$n \cdot Prod.[\,] = 8$

$= \{$ Def Prod $\}$

$n \cdot 1 = 8$

$= \{$ neutro $\}$

$\boxed{n = 8}$

(CR) $gPin8.n.(x \triangleright xs) = \langle \exists as, bs : (x \triangleright xs) = as +\!+\, bs : n \cdot Prod.as = 8 \rangle$

$= \{$ lógica $\}$

$\langle \exists as, bs : (x \triangleright xs) = as +\!+\, bs \wedge (as = [\,] \vee as \neq [\,]) : n \cdot Prod.as = 8 \rangle$

$= \{$ Distributiva, Partición de rango $\}$

$\langle \exists as, bs : (x \triangleright xs) = as +\!+\, bs \wedge as = [\,] : n \cdot Prod.as = 8 \rangle \vee \langle \exists as, bs : (x \triangleright xs) = as +\!+\, bs \wedge as \neq [\,] : n \cdot Prod.as = 8 \rangle$

$= \{$ eliminación de variable, Def Concat $\}$

$\langle \exists as, bs : (x \triangleright xs) = bs : n \cdot Prod.[\,] = 8 \rangle \vee \langle \exists as, bs : (x \triangleright xs) = as +\!+\, bs \wedge as \neq [\,] : n \cdot Prod.as = 8 \rangle$

$= \{$ Def de Prod, termino constante $\}$

$n = 8 \vee \langle \exists as, bs : (x \triangleright xs) = as +\!+\, bs \wedge as \neq [\,] : n \cdot Prod.as = 8 \rangle$

$= \{$ Cambio de variable $as := (a \triangleright as)$, neutro del $\wedge$ $\}$

$\langle \exists a, as, bs : (x \triangleright xs) = (a \triangleright as) +\!+\, bs : n \cdot Prod.(a \triangleright as) = 8 \rangle$

$= \{$ Def Concatenar $\}$

$\langle \exists a, as, bs : (x \triangleright xs) = a \triangleright (as +\!+\, bs) : n \cdot Prod.(a \triangleright as) = 8 \rangle$

$= \{\text{Prop lista}\}$

$\langle \exists a, as, bs : \underline{x = a} \wedge xs = as ++ bs : n \cdot Prod.(a \triangleright as) = 8 \rangle$

$= \{\text{eliminación de variable}\}$

$\langle \exists a, as, bs : xs = as ++ bs : n \cdot \underline{Prod.(x \triangleright as)} = 8 \rangle$

$= \{\text{Def de Prod}\}$

$\langle \exists a, as, bs : xs = as ++ bs : \underline{n \cdot x} \cdot Prod.as = 8 \rangle$

$= \{\text{asociatividad}\}$

$\langle \exists as\ bs : xs = as ++ bs : (n \cdot x) \cdot Prod.as = 8 \rangle$

$= \{HI\}$ $gPin8.(n \cdot x).xs$

$Pin8 : [Num] \to Bool$

$Pin8.[\,] \doteq False$

$Pin8.(x \triangleright xs) \doteq False \vee gPin8.x.xs$

- $gPin8 : Num \to [Num] \to Bool$

$gPin8.n.[\,] \doteq False$

$gPin8.n.(x \triangleright xs) = n = 8 \vee gPin8(x \cdot n).xs$

- $gPin8.0.xs = Pin8.xs$

b) $Pin8.xs = \langle \exists as, bs : xs = as ++ bs : Prod.as = 8 \rangle$

$Pin8.[2,2,2] \doteq \langle \exists a, bs : [2,2,2] = as ++ bs : \underline{Prod.as = 8} \rangle$

as	bs
[ ]	[2,2,2]
[2]	[2,2]
[2,2]	[2]
[2,2,2]	[ ]

$= \{\text{Definición de rango}\}$

$\langle \exists as, bs : as \in \{[\,], [2], [2,2], [2,2,2]\} : Prod.as = 8 \rangle$

$= \{\text{Aplicación del rango al término}\}$

$\underline{Prod.[\,] = 8} \vee \underline{Prod.[2]} = 8 \vee \underline{Prod.[2,2]} = 8 \vee \underline{Prod.[2,2,2]} = 8$

$= \{\text{Def Producto}\}$

$1 = 8 \vee 2 = 8 \vee 4 = 8 \vee 8 = 8$

$= \{\text{lógica}\}$

$F \vee F \vee F \vee T$

$= \{\text{elemento absorbente}\}$

$|True|$

3. Especificar funciones para resolver los siguientes problemas. También dar el tipo. **No derivar.**

✓ a) Calcular la cantidad de divisores que tiene un número $n > 0$ (sin contar el 1 ni n mismo).

✓ b) Dadas dos listas xs e ys, calcular si xs tiene algún segmento cuya suma es mayor que la suma de ys. **Ejemplo:** con $xs = [-1, 8, -2, 4, -2]$, $ys = [3, 6]$ el resultado es afirmativo, por el segmento $[8, -2, 4]$ de xs.

a) $\langle \sum i : 0 < i < n \wedge i \bmod n = 0 : 1 \rangle$

b) $\langle \exists a, bs ; xs = as ++ bs \wedge \langle \forall cs, ds : ys = cs ++ ds : sum.ys \rangle : sum\,as \geq sum.ys \rangle$